Classical inference at scale

dark energy, tensions, and the GPU revolution

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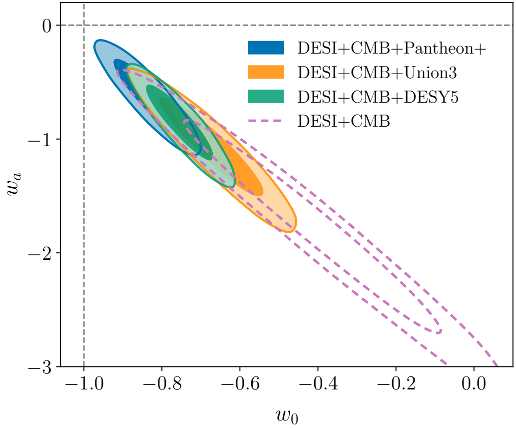
DESI DR2: evolving dark energy?

- DESI DR2: frequentist analysis reports 4.2σ preference for $w_0 w_a$CDM over ΛCDM [2503.14738].
- DES-Dovekie recalibration [2511.07517] [2506.05471]: reduces to 3.2σ.
- We ran a comprehensive Bayesian reanalysis: 248 dataset–model combinations via nested sampling.
- The signal is driven by inter-dataset tension, not coherent dark-energy evolution.

This talk: the evidence, the GPU tools that will take it further, and how we build them fast.



The three pillars of (Bayesian) inference

Parameter estimation

What do the data tell us about the parameters of a model? *e.g. the size or age of a ΛCDM universe*

$$P(\theta|D,M) = \frac{P(D|\theta,M)P(\theta|M)}{P(D|M)}$$

$$\mathcal{P} = \frac{\mathcal{L}\times\pi}{\mathcal{Z}}$$

$$\text{Posterior} = \frac{\text{Likelihood}\times\text{Prior}}{\text{Evidence}}$$

Model comparison

How much does the data support a particular model? *e.g. ΛCDM vs a dynamic dark energy cosmology*

$$P(M|D) = \frac{P(D|M)P(M)}{P(D)}$$

$$\frac{\mathcal{Z}_M\Pi_M}{\sum_m Z_m\Pi_m}$$

$$\text{Posterior} = \frac{\text{Evidence}\times\text{Prior}}{\text{Normalisation}}$$

Tension quantification

Do different datasets make consistent predictions from the same model? *e.g. CMB vs Type IA supernovae data*

$$\mathcal{R} = \frac{\mathcal{Z}_{AB}}{\mathcal{Z}_A\mathcal{Z}_\mathcal{B}}$$

$$\log\mathcal{S} = \langle\log\mathcal{L}_{AB}\rangle_{\mathcal{P}_{AB}} - \langle\log\mathcal{L}_A\rangle_{\mathcal{P}_A} - \langle\log\mathcal{L}_B\rangle_{\mathcal{P}_B}$$

[1902.04029] [1903.06682] [1910.07820]

unimpeded: PLA for the next generation

- DiRAC allocations: 30M + 20M CPU-hours (DP192, DP264).
- Main goal: Planck Legacy Archive equivalent:

  Param. est. → Model comparison
  MCMC → Nested sampling
  Planck → {Planck, DES, DESI, BAO, ...}
- All pairwise and triplet combinations.
- Python library: download, cache, analyse with `anesthetic`. [1905.04768]

github.com/handley-lab/unimpeded

PolyChord: [1502.01856] [1506.00171]

unimpeded: DESI DR2 reanalysis

248 nested-sampling runs across 8 models and 11 datasets

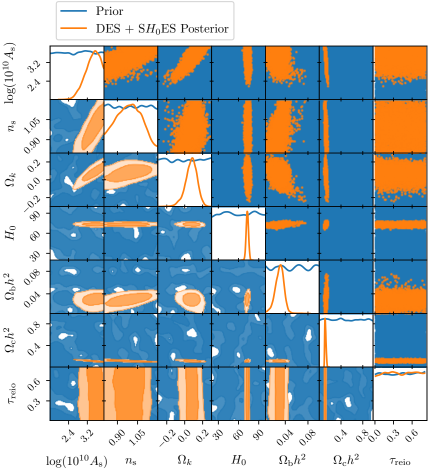
- 248 runs: 11 datasets × 8 models, plus all pairwise and triplet combinations.
- Datasets: Planck, DESI DR1/DR2, DES-Y5, Pantheon$^+$, Union3, ACT, SPT, SH0ES.
- Models: ΛCDM, wCDM, $w_0 w_a$CDM, m_ν, Ω_k, r, A_L, $n_{\rm run}$.

[2511.05470] Nested sampling database

[2511.04661] Model comparison & tension grid

[2511.10631] Bayesian evidence for evolving DE

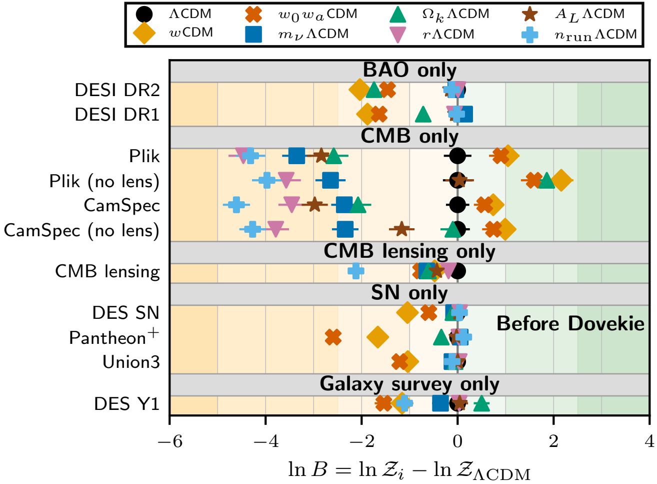
Ong, Yallup & Handley Full DESI DR2 reanalysis (coming soon)



Model posteriors: individual probes

$\ln B = \ln \mathcal{Z}_i - \ln \mathcal{Z}_{\Lambda\mathrm{CDM}}$ for 8 models across individual datasets

- DESI BAO penalises dynamic DE; $m_\nu\Lambda$CDM remains competitive.
- Planck CMB mildly prefers extensions ($\Delta \ln B \approx +1$ for $w_0 w_a$CDM).
- Supernovae alone lack discriminatory power.
- Opposing single-probe preferences set up the dataset combination story.



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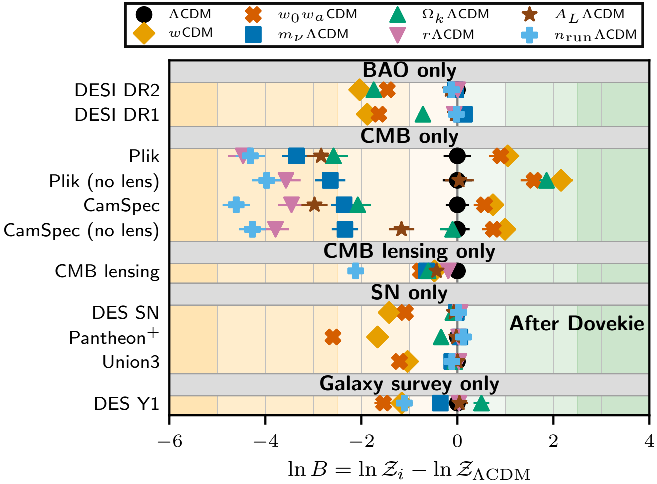
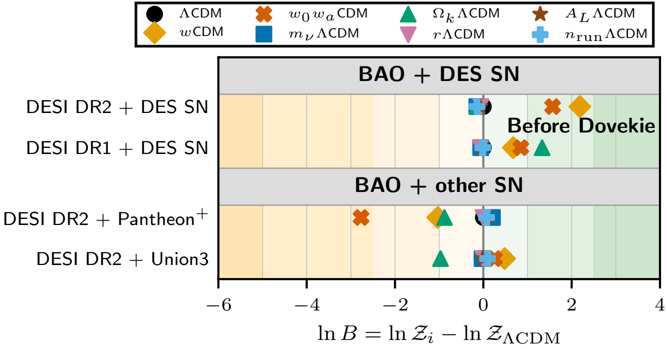


Model posteriors: pairwise combinations

$\ln B = \ln \mathcal{Z}_i - \ln \mathcal{Z}_{\Lambda\mathrm{CDM}}$ **for 8 models across pairwise datasets**

- BAO + CMB: $w_0 w_a$CDM suppressed ($\ln B = -1.22$), ΛCDM preferred.
- BAO + DES-Y5: preference shifts to dynamic DE.
- BAO + Pantheon$^+$: ΛCDM locked in.
- The SN catalogue choice is decisive.



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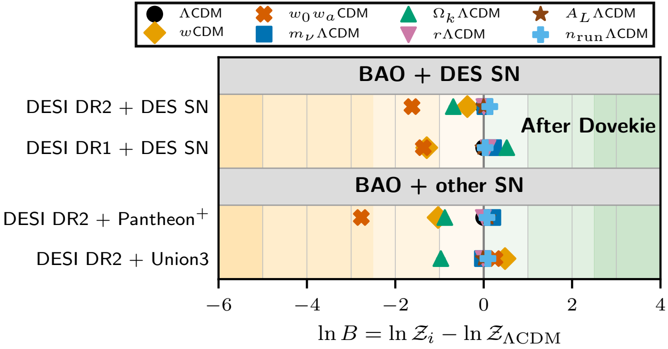
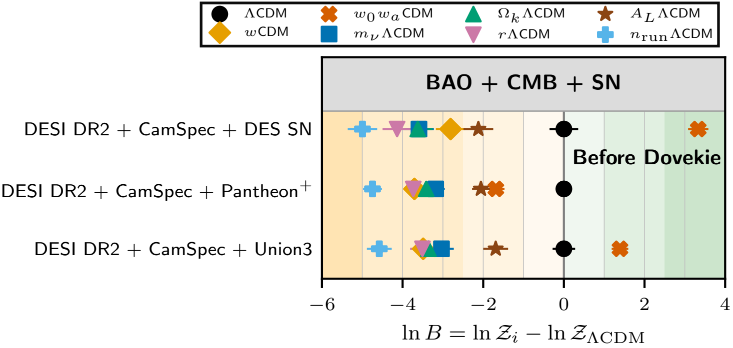


Model posteriors: triplet combinations

$\ln B = \ln \mathcal{Z}_i - \ln \mathcal{Z}_{\Lambda\mathrm{CDM}}$ for 8 models across BAO+CMB+SN

- BAO + CMB + DES-Y5: $w_0 w_a$CDM strongly preferred ($\Delta \ln B \approx +3.3$, $\sim 3\sigma$).
- BAO + CMB + Pantheon$^+$: ΛCDM holds ($\ln B = -0.38$ vs -2.06).
- The claim for new physics is entirely contingent on a single SN catalogue.



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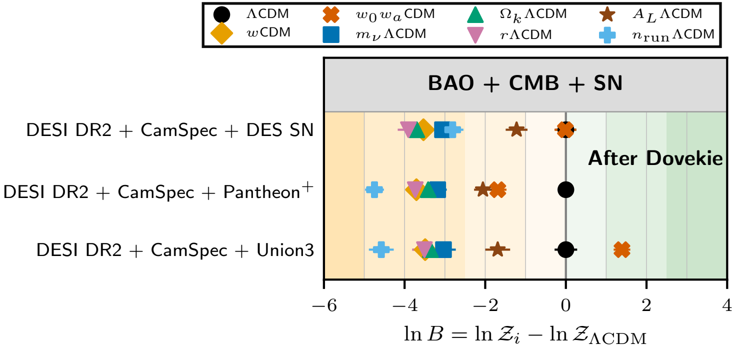
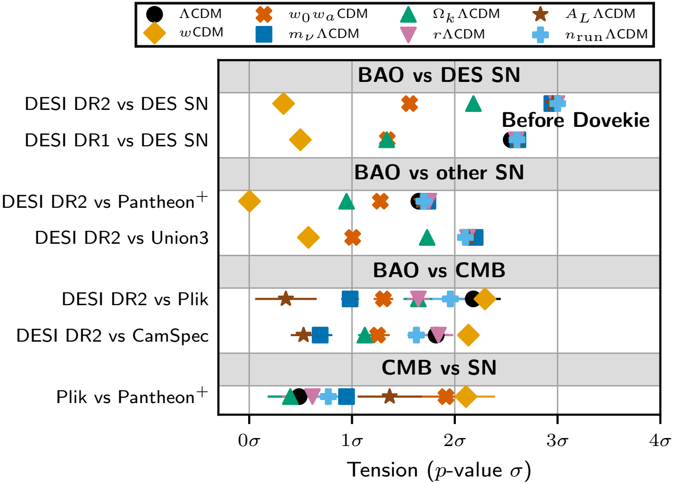


Where does the signal come from?

Pairwise tension significance (σ) across 8 models

- DESI DR2 vs DES-Y5 in ΛCDM: $2.95 \pm 0.04\,\sigma$.
- In wCDM: $0.33\,\sigma$. In $w_0 w_a$CDM: $1.56\,\sigma$.
- Extended models *resolve* the tension rather than detect new physics.
- Look-elsewhere threshold for $N = 200$ tests: $\sigma > 2.88$.



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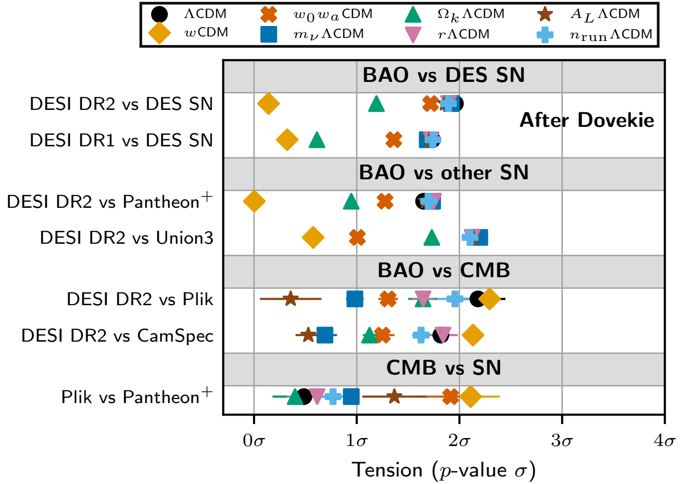


Why 4.2σ doesn't settle it

The Jeffreys–Lindley paradox

Frequentist ($\Delta\chi^2$)

- Rejects the null ($w = -1$) when the best fit deviates *at all*.
- With enough data, *any* exact null is rejected — $p \to 0$.
- 4.2σ: "the data are unlikely under ΛCDM."

Bayesian (Bayes factor)

- Compares full models: ΛCDM (N params) vs $w_0 w_a$CDM ($N+2$).
- Extra parameters pay an Occam penalty — prior volume wasted on bad regions.
- Unless the data *concentrate* away from $w = -1$, ΛCDM wins.

The paradox

As data grow, the two frameworks diverge: arbitrarily significant p-values coexist with Bayes factors favouring the null. DESI's 3.2–4.2σ is exactly this regime.

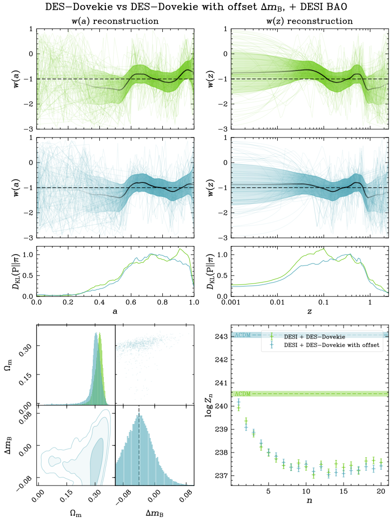
Beyond $w_0 w_a$: flexible dark energy

Independent confirmation from flexknot reconstruction

Flexknot $w(a)$ reconstruction [2503.17342] [2503.08658] [2509.13220]

- $w_0 w_a$ is too restrictive — flexknots let the data speak.
- W-shaped structure in $w(a)$; DESI and DES-5Y are in tension.
- A magnitude offset Δm_B between low-z and DES SNe beats flexknot DE in Bayesian evidence.

20 model complexities × multiple datasets — only feasible because the pipeline is in JAX: ***days*** *→* ***minutes****.*



Two revolutions

Distinct shifts, often conflated

1. GPUs for analysis

- GPU hardware accelerates *classical* statistical methods
- Gravitational Waves, Dark Energy, Supernovae, Radio
- The method transfers across astronomy

2. LLMs for development

- How we built this in months, not years — and we're just getting started
- Verification, legacy code, and new workflows
- Expanding the scope of what a small group can tackle

3. Live Demo & Closing

Loop running in front of an audience: the most dangerous game.

The frontier of inference

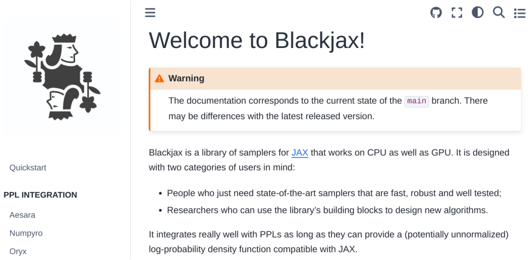
What becomes possible with GPU acceleration

- These analyses work, but they're slow. If we can speed them up, new science becomes possible:

  Robustness Marginalise over 100+ nuisance parameters rather than fixing them.

  Discovery at scale Definitive Bayesian evidence for new physics across grids of models and datasets.

  Low latency Inference fast enough for triggers, scheduling, and real-time follow-up.



- Nested Sampling on a GPU [2601.23252]
- Alongside MCMC, HMC, SMC [2402.10797]
- In Google's premier GPU language
- *No gradients required.*

Real-time gravitational wave PE

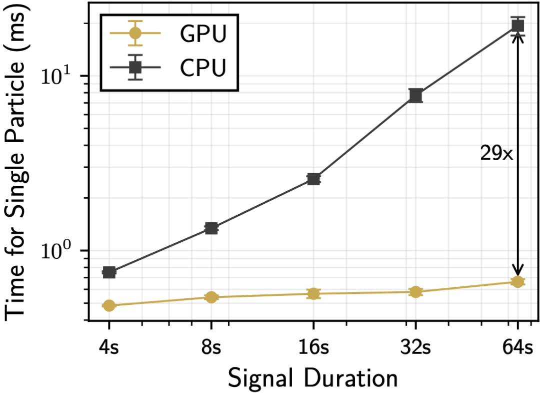
From 2 hours on 60 CPUs to 30 seconds on one GPU

The problem [1710.05833]

- GW170817: binary neutron star merger. EM follow-up needed fast sky localisation.
- Full parameter estimation (`bilby`): **60 CPUs, 2 hours**. Too slow for transients.

GPU nested sampling [2509.04336] [2509.24949] [2601.21630]

- **1 GH200 GPU**, **∼30 seconds**. 4 s of real LIGO data (GW150914), full 15-D parameter space.
- Near-real-time: EM alerts while the event is still physically interesting.



Extending to the CMB

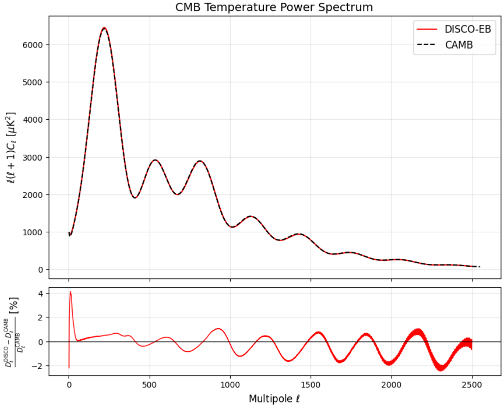
The missing piece for GPU-native cosmological inference

The missing component

- BAO and SNe are in JAX. The CMB is not — CAMB/CLASS are legacy Fortran/C.
- To fold in Planck/ACT properly, we need a GPU-native Boltzmann solver.

DISCO: GPU-native Boltzmann solver

- Full Boltzmann solver in JAX, GPU-native, auto-differentiable.
- Sub-percent agreement with CAMB (0.7% mean residual).
- $\sim$5 s per model on H100 and falling; saturates well on a GPU.



GPU-accelerated supernova photometry

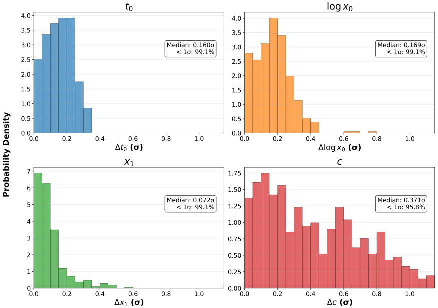
Supernovae as standard candles

`jax-bandflux`: SN photometry on GPU [2504.08081]

- JAX SALT3 bandflux integration $\Rightarrow$ $\sim$**100**$\times$ **speedup**, fully differentiable.

Bayesian anomaly detection [2509.13394]

- Each data point gets a latent anomaly probability, fitted jointly with SALT3 parameters.
- Automates outlier flagging, filter rejection, *and* data preservation within partially contaminated bands.
- 98% of parameters match manual analysis — but fully automated for LSST.

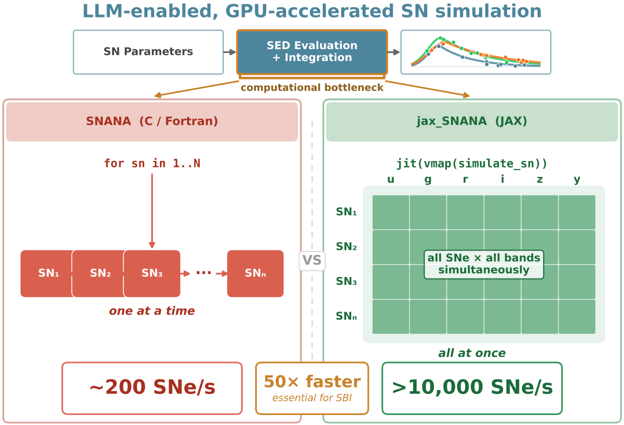


Simulations at scale

Supernovae as standard candles

JAX-SNANA: 10,000 sims/sec on A100

- Full SN light-curve simulation pipeline rewritten in JAX.
- $\sim$200 sims/sec (CPU) $\rightarrow$ $>$**10,000/sec** (GPU): 50$\times$ speedup.
- Validated against DES Y5 to <1 millimag. Essential for SBI at LSST scale.

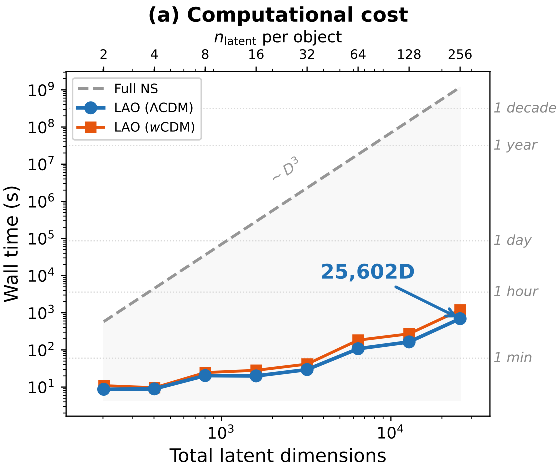


Laplace Approximation Optimisation

- SALT2 standardises with 2 parameters per SN. A physical model could use 100s of measurements per object.
- **25,602-D** likelihood in **11 minutes**. NS: $D^3 \rightarrow$ a decade of H200 time.

Autodiff for marginalisation [2509.13307]

- Optimise to mode, auto-diff the Hessian, marginalise exactly. Parallelised across objects on GPU.
- No manual derivations — auto-diffs the forward model. Change the physics, re-run.



Marginalising latent variables at scale

Supernovae as standard candles

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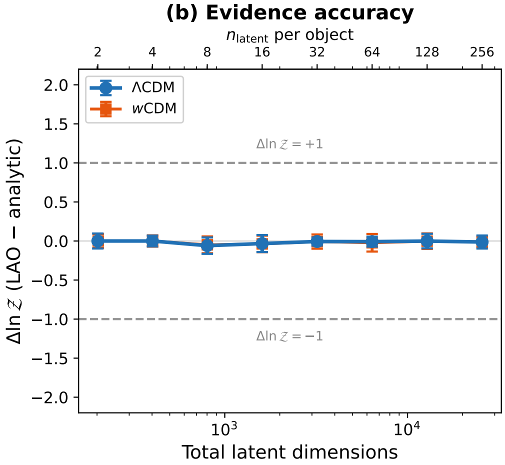


A brief history

How we got here, fast

2019 Tabnine — deep learning code completion

Jun 2021 GitHub Copilot — AI pair programming

Nov 2022 ChatGPT (GPT-3.5) — "the moment"

Mar 2023 GPT-4 — qualitative leap

Dec 2023 Gemini 1.0 — Google enters the race

Jun 2024 Claude 3.5 Sonnet — the coding model

Sep 2024 OpenAI o1 — reasoning models

Oct 2024 Cursor — agentic IDE

Jan 2025 DeepSeek R1 — open-source reasoning

May 2025 Claude Code — agentic coding CLI

Aug 2025 GPT-5 — unified model

Nov 2025 Gemini 3, Opus 4.5

Dec 2025 GPT-5.2

2025– Qwen, Kimi, GPT-OSS — open source catches up

Feb 2026 Opus 4.6 — *this talk*

Three ways AI meets science

Only one matches what science actually is

Foundation Models

- "Train a model on all astronomical data."
- Expensive, opaque, unclear payoff.
- Physics is about *compression*, not prediction.

Autonomous Agents

- "Click a button, generate a paper."
- Produces "science-like" content without understanding.
- Science is a human culture of critique — not a pipeline.

Augmented Humans

- Humans accelerated, not replaced.
- Humans define the physics and verification; AI handles the syntax.
- **Our approach.**

"Silicon Valley thinks a scientist is something that takes in coffee and produces fusion reactors. It's not."

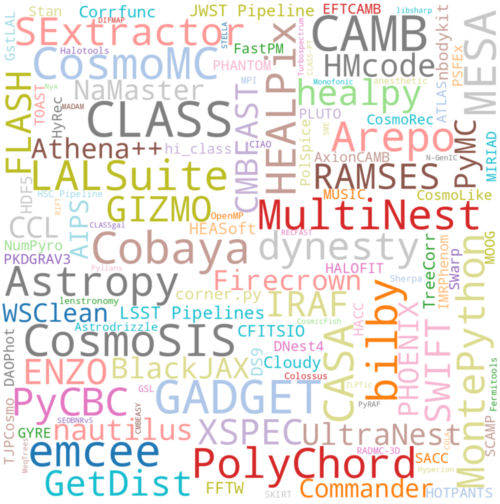
The code debt problem

Why PhD students spend years operating frameworks instead of doing physics

- Our fields are riven with **code debt**.
- Students end up *operating* CAMB, CLASS, Cobaya — not implementing their own ideas.
- We mistake “ability to debug legacy Fortran” for “scientific aptitude.”

The shift

LLMs are autocomplete on steroids — our route out of the spiral. We move from **maintaining code** to **specifying physics**.



The mechanism: context engineering

From “prompting” to “orchestration”

2025: Prompt engineering

- *“Write me a Python loop.”*
- Limited by user memory and syntax knowledge.
- Useful for snippets, bad for systems.

2026: Context engineering

- *“Here is the CAMB repo. Find the neutrino integration logic.”*
- LLMs acting on the **full context** of the project.
- The bottleneck is no longer generating code, but **evaluating** it.

Development time collapsed

The work didn't disappear — it moved to verification

Implementation `emcee` in BlackJAX: 2 months (manual) → **15 minutes** (LLM).

Optimisation Natalie Hogg: covariance matrix 24 h → **1–2 min**. Vectorisation, JIT, quasi-Monte Carlo.

Portability Sam Leeney: REACH anomaly detection → supernova cosmology in hours. [2504.08081]

New science GW emulator with James Alvey: nothing → working package in **8 hours**.

The insight

I mistook *mental athleticism* — typing speed, syntax memory, algebraic calculation — for scientific strength.

With AI, **scientific principles** and **creative drive** become the differentiators.

"We didn't feel tired — we were spending all our time reasoning about physics."

Verification is the new lab bench

LLMs make wrong code easy. Testing makes it detectable.

Reference outputs

Match legacy code to floating-point precision.
Regression to known-good results.

Simulation-based calibration

KS-tests on samples (SBC).
Statistical validation of the full pipeline.

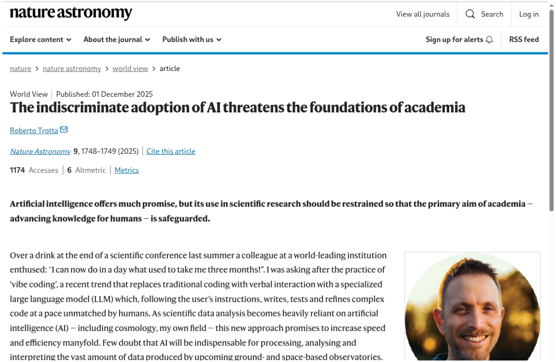
Independent testing

One person implements, another writes tests.
Adversarial review: one LLM writes, another critiques.

"We are no longer limited by how fast we write code — we are limited by how rigorously we verify it."

Two serious arguments

Both from February 2026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World View | Published: 01 December 2025

The indiscriminate adoption of AI threatens the foundations of academia

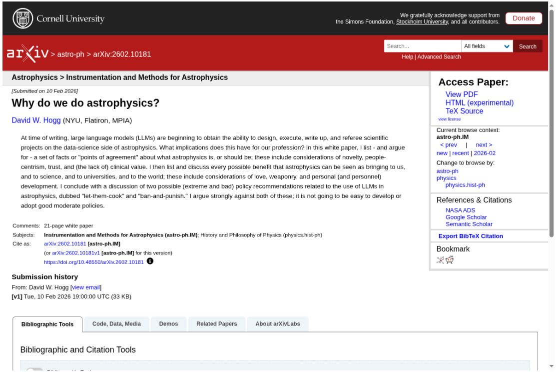
Roberto Trotta

Nature Astronomy 9, 1748–1749 (2025) | Cite this article

1174 Accesses | 6 Altmetric | Metrics

Artificial intelligence offers much promise, but its use in scientific research should be restrained so that the primary aim of academia – advancing knowledge for humans – is safeguarded.

Over a drink at the end of a scientific conference last summer a colleague at a world-leading institution enthused: "I can now do in a day what used to take me three months!". I was asking after the practice of 'vibe coding', a recent trend that replaces traditional coding with verbal interaction with a specialized large language model (LLM) which, following the user's instructions, writes, tests and refines complex code at a pace unmatched by humans. As scientific data analysis becomes heavily reliant on artificial intelligence (AI) – including cosmology, my own field – this new approach promises to increase speed and efficiency manyfold. Few doubt that AI will be indispensable for processing, analysing and interpreting the vast amount of data produced by upcoming ground- and space-based observatories.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arXiv > astro-ph > arXiv:2602.10181
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Astrophysics > Instrumentation and Methods for Astrophysics
[Submitted on 10 Feb 2026]

Why do we do astrophysics?

David W. Hogg (NYU, Flatiron, MPIA)

At time of writing, large language models (LLMs) are beginning to obtain the ability to design, execute, write up, and referee scientific projects on the data-science side of astrophysics. What implications does this have for our profession? In this white paper, I list - and argue for - a set of facts or "points of agreement" about what astrophysics is, or should be; these include considerations of novelty, people-centrism, trust, and (the lack of) clinical value. I then list and discuss every possible benefit that astrophysics can be seen as bringing to us, and to science, and to universities, and to the world; these include considerations of love, weaponry, and personal (and personnel) development. I conclude with a discussion of two possible (extreme and bad) policy recommendations related to the use of LLMs in astrophysics, dubbed "let-them-cook" and "ban-and-punish." I argue strongly against both of these; it is not going to be easy to develop or adopt good moderate policies.

Comments: 21-page white paper
Subjects: **Instrumentation and Methods for Astrophysics (astro-ph.IM)**; History and Philosophy of Physics (physics.hist-ph)
Cite as: arXiv:2602.10181 **[astro-ph.IM]**
(or arXiv:2602.10181v1 **[astro-ph.IM]** for this version)
https://doi.org/10.48550/arXiv.2602.10181

Submission history
From: David W. Hogg [view email]
[v1] Tue, 10 Feb 2026 19:00:00 UTC (33 KB)

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Two serious arguments

Both from February 2026

Trotta [2602.10165]

- LLM reliance causes measurable cognitive decline (Kosmyna et al. 2025).
- Scientific coding requires understanding the models — outsourcing this undermines science.
- Students risk becoming "little more than prompt engineers."

Hogg [2602.10181]

- Students are ends, not means — "better than a grad student" is a dangerous framing.
- LLMs cannot take responsibility or be authors.
- Neither "let them cook" nor "ban and punish" works — no clear middle way.

My response

I agree with both. Use LLMs for overhead, not for the science. Verify rigorously.

I write talks without typing

Case study: IoA Colloquium, February 2026

What I used

- **Voice**: Otter.ai transcription → notes
- **Loop runner**: Claude Code (Opus 4.6)
- **Reviewers**: Gemini 3 and GPT-5.2 as independent critics
- **Knowledge base**: 1 year of 450 transcripts, whiteboard photos, emails, LaTeX of papers and grants — in a context management system
- **Email**: 14 personalised figure requests, sent and tracked
- **Web**: Playwright, arXiv, Wikimedia Commons
- **Drafting**: Gemini (1M context) for slides from source material

What I did

- Talked about the science
- Chose the figures
- Edited the emails
- Said "no" a lot
- Verified the physics

What I didn't do

Type. Read Fortran. Manually search 450 transcripts. Write 14 emails from scratch. Make a single LaTeX file by hand.

Live Demo

The generate → test → iterate loop, not one-shot codegen

[Claude Code Demo]

Conclusions

GPUs $\neq$ AI

- Classical statistics on GPU is competitive with ML — and interpretable.
- GW PE: 2 h $\rightarrow$ 30 s. Dark energy: days $\rightarrow$ minutes. Supernovae: 50$\times$. The method transfers across astronomy.

LLMs $\neq$ science

- LLMs don't replace scientists — they make code and text virtually free.
- The bottleneck shifts from *writing* to *verifying*. Verification is a higher-order scientific skill.

Seize the opportunity

Those who combine rigorous inference with LLM-accelerated development will lead the next decade of astronomy. Capture content. Engineer context. Stay the scientist.

If producing code and text were **virtually free***, what would you do differen*

handley-lab.co.uk [github:handley-lab]